

Energy and Exergy Analysis of Solar Drying System

*A Mini-Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

IN

MECHANICAL ENGINEERING

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April, 2026



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CERTIFICATE

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Acknowledgements

The satisfaction that accompanies the successful completion of the task would be incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Dr. A.C. Uma Maheswer Rao**, Assistant Professorproject supervisor, Department of Mechanical Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Dr. P. Venkateshwar Reddy**, the head of the Department, Department of Mechanical Engineering, his guidance, intense support, and encouragement, which helped us to mold our project into a successful one.

We show gratitude to our honorable principal, **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman, and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of the Electronics and Communication Engineering department for their valuable support and generous advice. Finally, thanks to all our friends and family members for their continuous support and enthusiastic help.

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Abstract

The objective of this research is to explore the performance and exergy analysis of an indirect solar dryer (ISD) for drying plums in natural and forced convection modes. The research is aimed at improving the performance and evaluation of the thermodynamic performance of the solar drying system. The ISD system consists of a solar air collector, a drying chamber, an air duct, and an airflow control mechanism to supply hot air for removing moisture content from plums. The analysis was carried out to determine the efficiency of the collector, the efficiency of the dryer, the inflow and outflow, and the exergy loss. The results showed that the collector and drying efficiencies are higher for forced convection mode compared to natural convection mode. The analysis also showed that the solar collector experienced moderate exergy loss because of thermal irreversibility, while the drying chamber used a high rate of exergy for the drying process. The results also showed that the exergy efficiency of the drying chamber was significantly higher than that of the solar collector, indicating that exergy was effectively used for evaporating moisture content from plums. The results also showed that factors such as the sustainability index, waste exergy ratio, and improvement potential are important for evaluating the thermodynamic performance of the system. The results showed that forced convection is a better method for improving the performance and efficiency of the ISD compared to natural convection mode for drying plums. In conclusion, the results showed that the use of an indirect solar dryer is an efficient and sustainable method for preserving plums while reducing energy loss and improving the efficiency and quality of the products.

Keywords: Indirect Solar Dryer (ISD) ; Solar Air Collector (SAC)

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Abbreviations

Abbreviation	Description
ISD	Indirect Solar Dryer
SAC	Solar Air Collector
PCM	Phase Change Material
PV	Photovoltaic
DC	Drying Chamber

CHAPTER 1

Introduction

1.1 Introduction

One of the most common ways of preserving agricultural produce is drying, which helps to lower the level of moisture and growth of microbes. In most of the rural locations, traditional open-sun drying is commonly applied because it is simple and quite cheap. This process, however, has a number of drawbacks that include being contaminated by dust and insects, not being determined by weather, drying unevenly, and loss of product quality. In order to address these shortcomings, solar drying technologies have been invented to offer a controlled drying environment and use renewable energy. Solar drying enhances the efficiency of drying in addition to minimising losses after harvest and retaining higher quality products than the traditional forms of drying. The project will deal with the design and performance of an adapted Indirect Solar Dryer (ISD) in drying plums. The system comprises a solar air collector and a drying chamber, whereby the solar energy is transformed into a thermal energy that heats the air that is then put into the drying chamber. Some of the adjustments made in the system design include adding some changes aimed at improving the thermal performance of the system. They are incandescent bulbs fitted on the solar collector to add more heat, heating coils installed on the edges of the blower fans to help increase air temperature, copper tubes loaded with phase change material (PCM) wax to serve as thermal storage capacity, and aluminum foil coating the inside of the drying chamber to enhance better heat reflection and reduce heat loss. Moreover, the forced air circulation is achieved through the application of the blower fans, and the photovoltaic (PV) panel is employed to operate the blowers, making the system more efficient in terms of energy consumption and environmentally friendly. This research is also important because it may produce an effective and sustainable drying

system based on the use of renewable energy resources. Agricultural products like plums contain a lot of moisture, and thus, when not well preserved, they spoil easily. Through the thermal enhancement and integration of renewable energy, the proposed system will be able to reduce post-harvest losses as well as enhance product quality alongside offering an environmentally friendly food preservation option, especially in rural and agricultural areas, where traditional energy sources might be scarce. The overall objective of this project will be to test the drying efficiency and thermodynamic efficiency of the modified indirect solar dryer. The research includes the analysis of the system in various conditions during operation and energy and exergy analysis to identify such significant parameters as energy efficiency, exergy efficiency, exergy losses, sustainability index, improvement potential, and the waste exergy ratio. In this analysis, the project will evaluate the efficiency of the added components in enhancing the performance of the system in drying.

1.2 Background and Motivation

In this section, the background that has resulted in the project development is presented. It outlines the area of solar drying technology and the difficulties that come with the conventional drying processes that are applied to farm products. Although traditional open sun drying has been used widely, there are several issues associated with it which include contamination, uneven drying, and reliance on weather conditions. Past studies have analyzed different enhancement in solar drying system such as adoption of solar air collector, forced convection air movement, storing thermal energy in the system, and thermodynamic evaluation to enhance drying efficiency and produce good products.

This project was motivated by the fact that the development of an efficient and reliable solar drying system that can eliminate the constraints of the traditional systems is needed. The project would upgrade the performance and sustainability of solar dryers by adding further features like supplementary heating systems, thermal storage in the form of phase change material (PCM), better ways of retaining heat, and airflow systems which would be powered

by renewable energy. The background information assists in providing the context in which the study is located, whereas the motivation outlines the significance and the need to come up with better solar drying technologies to be implemented in agriculture.

The current project is located in the sphere of renewable energy engineering and agricultural drying technology, specifically the sphere of solar thermal system in food preservation. One of the techniques that have been utilized in the drying of agricultural products using solar energy is solar drying. It is extensively used in the post harvest processing so as to curb spoilage, preserve the quality of the product in addition to increasing the shelf life of the food commodities of fruits, vegetable as well as grains. Due to the growing need of sustainable and environmentally friendly technologies in the world economy, solar drying systems have become a significant field of research in renewable energy usage. The combination of solar air collector, thermal storage of energy, photovoltaic systems, and thermodynamic analysis of performance contribute to even more effective construction of solar drying systems. In the past, agricultural goods were conserved simply by sun drying, in which crops are laid in the open sun to dry to minimize the amount of moisture they have. This technique is an ancient one because it is not complex or expensive. Nevertheless, open sun drying is limited in a number of ways such as exposure to dust, insects and animals, and reliance on weather conditions and control over the drying temperature and airflow. Such problems normally result in uneven drying, contamination and low quality of products.

In order to address these constraints, scientists started to work on the creation of solar drying technologies offering to a greater extent controlled drying conditions. In the course of time, a number of solar dryers have been developed and they are direct solar dryers, indirect solar dryers and mixed mode solar dryers. The Indirect Solar Dryer (ISD) has received much attention among them owing to the fact that the product does not come into direct contact with solar radiation. A solar air collector is instead used to absorb the solar energy, and it then heats the air entering the drying chamber. Moisture in the product is removed by the convection of the product using the hot air

and enhances the efficiency of drying and the quality of the product.

The inventions of technology in the recent years have enhanced solar drying systems. To improve the performance of the systems and ensure the drying temperatures remain steady, researchers have proposed forced convection airflow with the use of blower fans, thermal energy storage systems, phase change materials (PCM) and auxiliary heating sources. These advances have seen the solar dryers become more dependable and effective especially in areas where the solar radiation can vary throughout the drying session. Some scientists have also worked on enhancing thermal performance of solar air collectors which form major constituents of solar drying systems. It has been increased by making changes in the form of better absorber plates, better airflow channels, and better absorbing material used in insulation.

The other area in which research has been done is the application of phase change materials (PCM) in solar dryers. PCM has the ability of absorbing excess heat when the sun is bright and releasing the stored heat when the sun is not bright. This ensures that it has a more stable temperature of drying and is also capable of carrying on the drying process during cloudy weather or when the sun sets.

Moreover, energy and exergy analyses have been extensively used in the field of solar drying to assess the thermodynamic work of system of drying. The efficiency of the system to use solar heat is measured with energy analysis and losses in energy and irreversibilities in the system are measured with exergy analysis. These are analytical techniques that can assist researchers maximise the design and functioning of solar drying systems.

Irrespective of these developments, the need to establish superior solar drying systems that integrate various thermal enhancement practices with integration of renewable energy still exists. Thus, the project under consideration is aimed at an indirect solar dryer modified to use incandescent bulbs, heating coils, PCM-filled copper tubes, aluminum foil lining, blower fans, and a photovoltaic panel to improve the drying efficiency and energy efficiency. This solution will be intended to deliver a more stable, effective, and sustainable drying option to agricultural products like plums. The significance of

this project is that it aims at designing an effective and efficient solar drying system to dry agricultural produce that will enable to minimize the losses that occur at the end of harvest as well as give an opportunity to preserve the food. Most fruits like plums are very moist and they can easily spoil without proper preservations. Traditional open sun drying is still very common in several rural and agricultural regions where they are associated with numerous issues that include dust and insect contamination, uneven drying, poor quality of products and reliance on weather patterns. Consequently, it has been strongly demanded that better drying technologies should be adopted that will be able to offer controlled drying conditions and also make use of renewable energy.

The solar drying systems are also an ecologically friendly and efficient energy source since it uses solar energy as a source of heat. Traditional solar dryers however are at times limited by factors like temperature variances, low heat retention and the unpredictable airflow hence limiting the overall drying performance. It is necessary to overcome these shortcomings by ensuring that solar drying systems are more reliable.

The project will solve these issues by proposing an improved indirect solar dryer that has a number of thermal enhancement characteristics. The incandescent bulbs and heating coils are to make the device provide supplementary heating in the case of the lack of solar energy, and copper tubes filled with wax of phase change material (PCM) store some more heat and discharge it during a decrease in temperature. The aluminum foil lining on the inside of the drying chamber enhances the heating of the chamber by reflecting the thermal energy and the airflow is managed by blower fans that are supplied with renewable energy by a photo-voltaic panel. These are modifications to ensure that the drying temperatures remain constant, that the heat transfer is optimized and also that the rate of drying is accelerated.

Another significant research requirement that is met by the project is conducting the energy and exergy analysis to determine the thermodynamic efficiency of the system. Although numerous research works have examined the solar dryers, less detailed research has examined systems integrating various thermal enhancement methods with the renewable energy-driven airflow within

one drying unit. Through comparing the energy efficiency, exergy efficiency, exergy losses, sustainability index, improvement potential and waste exergy ratio, this paper will help in enhancing the understanding of the performance of advanced solar drying systems.

1.3 Objectives of the Project Work

List the primary objectives of the project. To design and develop a modified Indirect Solar Dryer (ISD) to dry plums with the help of the solar energy.

1. To improve the thermal efficiency of the solar air collector with the use of incandescent bulbs, heating coils, and copper tubes with phase change material (PCM) wax.

2. To increase the heat retention in the drying chamber with an aluminum foil lining to decrease the heat losses.

3. To achieve forced convection airflow with the help of blower fans that are driven by a photovoltaic (PV) panel.

4. To determine the performance of the system used to dry by studying the distribution of temperature and moisture removed during the drying process.

5. To conduct the analysis of the solar dryer in energy to find the efficiency of the solar air collector and drying chamber.

6. To perform exergy analysis in order to determine the thermodynamic performance in terms of exergy efficiency, exergy losses, sustainability index, potential to improve, and ratio of waste exergy. Break down the general goal into smaller, specific goals.

1. To come up with the design of the indirect solar dryer, the solar air collector and drying chamber.

2. To add other heating features like incandescent bulbs and heating coils into the solar collector to enhance the performance of air heating.

3. To add copper tubes stuffed with phase change material (PCM) wax to provide thermal energy storage in order to keep drying temperatures constant.

4. To enhance the thermal efficiency of the drying chamber by lining the interior with aluminum foil to reflect heat and minimise heat losses.

5. To add blower fans to allow forced convection airflow, to accomplish

increased heat transfer and moisture removal during drying.

6. To power the blower fans with photovoltaic (PV) panel in order to make the system run on renewable energy.

7. To measure and quantify the drying conditions including the temperature of the air and the moisture loss during the drying of plums.

8. To consider the efficiency and sustainability of the system by evaluating the overall performance of the dryer using energy and exergy analysis. Use clear, action-oriented language (e.g., “To develop,” “To analyze,” “To evaluate”). These objectives should align with solving the problem defined earlier.

The key aim of this research work is to examine the effects of incorporating multiple platforms into large-scale, cooperative wireless networks. The objectives of this work are set out below:

1. To design and analyse the relay selection criteria for faster data transmission in cooperative networking with multi channels.
2. To formulate an optimal solution for energy efficient transmission for large scale cooperative network with multiple channels.
3. To propose the optimal routing mechanism for large scale cooperative network by considering mobility factor.

1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter’s main points.

- Provide a brief description of what each chapter will cover.
- Typically, the structure would include:
 - **Chapter 2:** Literature Review
 - **Chapter 3:** Technical Title Related to the Prior Research

- **Chapter 4:** Technical Title Related to the Proposed Work
- **Chapter 5:** Results and Discussion
- **Chapter 6:** Conclusions and Recommendations

CHAPTER 2

Literature Review

2.1 Introduction

Solar drying technology has gained significant attention in recent years due to the increasing demand for renewable energy applications in agriculture and food processing. Conventional drying techniques such as open sun drying are widely used in many parts of the world, especially in developing countries. Although these methods are simple and inexpensive, they suffer from several disadvantages including contamination, uneven drying, dependence on weather conditions, and poor product quality. As a result, researchers have developed improved drying techniques that utilize solar energy in a controlled and efficient manner.

Solar dryers are designed to utilize solar radiation to generate heat, which is used to remove moisture from agricultural products. Compared to conventional drying systems that rely on fossil fuels or electricity, solar dryers offer several advantages such as reduced energy consumption, environmental sustainability, and lower operating costs. Over the years, numerous studies have been conducted to improve the design, performance, and efficiency of solar drying systems.

This chapter presents a review of previous research related to solar drying technologies and thermodynamic analysis methods. The literature review focuses on different types of solar dryers, improvements in solar air collectors, the use of thermal energy storage materials, and the application of energy and exergy analysis in evaluating solar drying systems. Understanding previous research helps identify current limitations and provides a foundation for developing improved solar drying systems.

2.2 Solar Drying Technologies

Solar drying is one of the oldest methods used for preserving agricultural products. It involves removing moisture from food products by using solar energy as the primary heat source. The drying process reduces the moisture content of the product, which prevents microbial growth and increases the shelf life of the product.

Solar dryers can be classified into three main types that are commonly used in agricultural applications.

2.2.1 Direct Solar Dryers

In direct solar dryers, the agricultural product is directly exposed to solar radiation. The product absorbs solar heat and moisture evaporates from its surface. Although direct solar dryers are simple and inexpensive, they may cause product discoloration, nutrient loss, and contamination due to direct exposure to sunlight.

2.2.2 Indirect Solar Dryers

Indirect solar dryers are widely used in research and industrial applications because the product is not directly exposed to solar radiation. In these systems, solar radiation is first absorbed by a solar air collector, which heats the air. The heated air is then passed into the drying chamber where it removes moisture from the product. This method improves product quality and provides better control over drying conditions.

2.2.3 Mixed-Mode Solar Dryers

Mixed-mode solar dryers combine the features of both direct and indirect drying systems. In this type of dryer, the product receives heat both from direct solar radiation and from heated air produced by the solar collector. Mixed-mode dryers generally provide higher drying rates compared to other solar drying systems.

Researchers have extensively studied these drying technologies to improve efficiency, reduce drying time, and enhance product quality.

2.3 Solar Air Collectors

The solar air collector is one of the most important components of a solar drying system. Its main function is to absorb solar radiation and convert it into thermal energy that heats the air used for drying.

A typical solar air collector consists of the following components:

- Transparent cover
- Absorber plate
- Air channel
- Insulation layer
- Frame structure

The transparent cover allows solar radiation to pass through and reach the absorber plate while minimizing heat loss. The absorber plate converts solar radiation into heat, which is then transferred to the air flowing through the collector.

Several researchers have focused on improving the efficiency of solar air collectors by modifying the absorber plate design, increasing surface area, improving insulation, and enhancing airflow patterns. These improvements help increase the air temperature and improve the overall drying performance of the solar dryer.

2.4 Energy Analysis of Solar Drying Systems

Energy analysis is commonly used to evaluate the performance of solar drying systems. It focuses on determining how efficiently solar energy is converted into useful thermal energy for the drying process.

In solar drying systems, energy analysis typically involves calculating:

- Useful heat gain
- Collector efficiency
- Drying efficiency
- Heat losses

The useful heat gain depends on factors such as solar radiation intensity, airflow rate, and temperature difference between inlet and outlet air. Energy efficiency indicates how effectively the solar collector converts solar radiation into heat energy.

Many researchers have used energy analysis to evaluate the performance of solar dryers and to identify ways to improve system efficiency.

2.5 Exergy Analysis of Solar Drying Systems

While energy analysis focuses on the quantity of energy, exergy analysis focuses on the **quality of energy** and the ability of that energy to perform useful work. Exergy analysis is a powerful thermodynamic tool used to identify inefficiencies and energy losses within a system.

In solar drying systems, exergy analysis helps determine:

- Exergy efficiency
- Exergy losses
- Sustainability index
- Improvement potential

Exergy losses occur due to irreversibilities in heat transfer processes, temperature differences, and environmental interactions. By identifying where these losses occur, researchers can improve the design and performance of solar drying systems.

Several studies have shown that exergy analysis provides deeper insights into system performance compared to energy analysis alone.

2.6 Thermal Energy Storage in Solar Dryers

One of the major challenges of solar drying systems is the variation in solar radiation during the day. When solar radiation decreases, the temperature inside the dryer may drop, which reduces the drying rate.

To overcome this problem, researchers have investigated the use of **thermal energy storage systems**. These systems store excess heat during periods of high solar radiation and release the stored heat when solar radiation decreases.

Phase Change Materials (PCM) are widely used for thermal energy storage in solar drying systems because they can store large amounts of heat during phase change and release it later. The use of PCM helps maintain stable drying temperatures and improves the overall efficiency of solar dryers.

2.7 Research Gap

Despite numerous studies carried out on solar drying technologies, several limitations still exist in current research. Most conventional solar drying systems rely solely on solar radiation as the primary heat source, making their performance highly dependent on weather conditions. During cloudy weather or periods of low solar radiation, the drying rate decreases significantly, resulting in longer drying times and reduced system efficiency.

Another limitation observed in previous studies is the lack of detailed thermodynamic analysis of solar drying systems. While many researchers have focused on energy efficiency, fewer studies have considered exergy analysis, which evaluates the quality of energy and identifies thermodynamic losses within the system. Without understanding exergy losses, it becomes difficult to determine where improvements can be made in system design.

In addition, many existing solar dryers suffer from heat losses due to poor insulation, inefficient collector designs, and improper airflow distribution. These losses reduce the overall efficiency of the drying system. Therefore, there is a need for further research that focuses on improving the thermal performance of solar drying systems while minimizing energy losses.

Another research gap relates to the optimization of system parameters such

as airflow rate, collector design, and drying chamber configuration. Many studies have not fully explored how these parameters influence the thermodynamic performance of solar drying systems.

Therefore, further investigation is required to analyze the performance of solar drying systems using both energy and exergy analysis in order to better understand system efficiency and identify areas for improvement.

2.8 Motivation of the Study

The motivation behind this study arises from the increasing demand for energy-efficient and environmentally friendly drying technologies in agricultural applications. Traditional drying methods often require large amounts of fossil fuel energy, which contributes to environmental pollution and increases operating costs.

Solar drying systems provide a sustainable alternative because they utilize renewable solar energy for the drying process. However, improving the efficiency of solar drying systems remains a challenge due to factors such as fluctuating solar radiation, heat losses, and inefficient system design.

This study is motivated by the need to improve the understanding of the thermodynamic performance of solar drying systems through detailed energy and exergy analysis. By analyzing both the quantity and quality of energy used in the drying process, it becomes possible to identify inefficiencies and develop strategies to improve system performance.

Furthermore, improving solar drying technologies can help farmers and food processing industries reduce post-harvest losses, improve product quality, and reduce dependence on conventional energy sources. Therefore, this research contributes to the development of sustainable food preservation technologies.

2.9 Objectives of the Study

The main objective of this project is to evaluate the performance of a solar drying system using energy and exergy analysis. The specific objectives of the study are as follows:

- To design and analyze a solar drying system for drying agricultural products using solar energy.
- To study the temperature variation in different parts of the solar dryer, including the solar air collector and drying chamber.
- To evaluate the drying performance of the system by measuring moisture removal and drying rate of the agricultural product.
- To perform energy analysis in order to determine the efficiency of solar energy utilization in the drying process.
- To conduct exergy analysis to identify thermodynamic losses and evaluate the quality of energy used in the system.
- To analyze the overall thermodynamic performance of the solar drying system and identify possible improvements for increasing system efficiency.

2.10 Summary

This chapter reviewed previous research related to solar drying systems and thermodynamic performance analysis. The literature shows that solar drying is an effective and sustainable method for preserving agricultural products. Various types of solar dryers, including direct, indirect, and mixed-mode dryers, have been developed to improve drying performance and product quality.

Researchers have also focused on improving the design of solar air collectors and integrating thermal energy storage systems to enhance drying efficiency. In addition, energy and exergy analyses have been widely used to evaluate the thermodynamic performance of solar drying systems and identify areas for improvement.

The literature review highlights the importance of combining energy and exergy analysis to better understand the efficiency of solar drying systems. These studies provide a strong foundation for the present project, which focuses on evaluating the performance of a solar drying system using thermodynamic analysis.

CHAPTER 3

Analysis of Solar Drying Energy and Exergy to Enhance its Efficiency

3.1 Introduction

Solar drying is a good technique of preserving agricultural products by harnessing renewable solar energy to remove moisture from food materials. Solar drying systems provide a cleaner, more energy-efficient, and environmentally friendly alternative for food preservation compared to conventional drying methods. However, the performance of solar dryers is influenced by several factors such as solar radiation intensity, airflow rate, temperature distribution, and system design. To improve the efficiency of solar drying systems, it is important to evaluate how effectively the available energy is utilized during the drying process.

Energy and exergy are thermodynamic tools commonly used to evaluate the performance of thermal systems. Energy analysis is used to estimate the amount of useful heat utilized in the drying process, whereas exergy analysis evaluates the quality of energy and identifies irreversibilities and energy losses within the system. The combined use of these approaches helps researchers better understand the effectiveness of solar drying systems and identify possible improvements.

This chapter discusses the application of energy and exergy analysis in solar drying systems and reviews previous studies related to improving solar dryer performance. The objective is to understand how thermodynamic analysis can be applied to increase the efficiency and reliability of solar drying technologies.

3.2 Solar Drying Systems

Solar dryers are designed to utilize solar radiation to heat air, which is then used to remove moisture from agricultural products. A typical solar drying system consists of three main components: a solar air collector, a drying chamber, and an airflow system.

The solar air collector absorbs solar radiation and converts it into thermal energy. The heated air generated in the collector is then directed into the drying chamber where it comes into contact with the product being dried. As the heated air flows over the product, moisture evaporates and is carried away by the airflow. This process reduces the moisture content of the product and helps extend its shelf life.

Solar dryers are generally classified into three types: direct solar dryers, indirect solar dryers, and mixed-mode solar dryers. Indirect solar dryers are commonly used in research because the product is not directly exposed to solar radiation, which helps preserve product quality and prevents contamination.

3.3 Energy Analysis of Solar Drying Systems

Energy analysis is used to evaluate how efficiently solar energy is converted into useful heat for the drying process. The energy efficiency of a solar dryer depends on several factors including solar radiation intensity, collector design, airflow rate, and heat losses within the system.

In a solar drying system, solar radiation absorbed by the collector increases the temperature of the air flowing through it. The heated air transfers heat to the product in the drying chamber, which causes moisture evaporation. Energy analysis helps determine how much of the available solar energy is effectively utilized for drying and how much energy is lost to the environment.

By evaluating the energy efficiency of different components of the system, such as the solar air collector and drying chamber, researchers can identify areas where improvements can be made to enhance system performance.

3.4 Exergy Analysis of Solar Drying Systems

While energy analysis focuses on the quantity of energy, exergy analysis evaluates the quality of energy and the useful work potential of the system. Exergy analysis helps identify irreversibilities and thermodynamic losses that occur during heat transfer and airflow processes.

In solar drying systems, exergy losses may occur due to temperature differences between system components, heat transfer inefficiencies, and environmental conditions. By analyzing these losses, researchers can determine how the system design can be improved to minimize energy degradation.

Important parameters used in exergy analysis include exergy efficiency, exergy loss, sustainability index, and improvement potential. These parameters help evaluate the thermodynamic performance of the solar drying system and provide guidance for optimizing the design and operation of the system.

3.5 Factors Influencing the Energy of Solar Drying

The efficiency of solar drying systems depends on several factors, including:

- **Solar radiation intensity**, which determines the amount of energy available for heating the air.
- **Airflow rate**, which influences heat transfer and the rate of moisture removal from the product.
- **Collector design and insulation**, which affect the amount of heat absorbed and the heat losses within the system.
- **Environmental conditions**, such as ambient temperature and humidity, which can influence the drying performance.

Improving these factors can significantly enhance the performance and reliability of solar drying systems.

3.6 Summary

This chapter discussed the role of energy and exergy analysis in improving the efficiency of solar drying systems. Solar dryers provide a sustainable and energy-efficient method for drying agricultural products, but their performance depends on several factors such as solar radiation, airflow, and system design.

Energy analysis helps determine how effectively solar energy is utilized during the drying process, while exergy analysis identifies thermodynamic losses and inefficiencies within the system. By applying both analyses, researchers can gain a deeper understanding of system performance and develop strategies for improving the efficiency of solar drying technologies.

The concepts presented in this chapter provide a foundation for evaluating the performance of the solar drying system used in this mini project and for identifying methods to enhance its efficiency.

CHAPTER 4

Experimental Setup and Methodology

4.1 Introduction

This chapter outlines the experimental setup and methodology used to evaluate the performance of the solar drying system in this mini project. The main objective of this study is to analyze how efficiently solar energy is utilized in drying agricultural products and to determine the thermodynamic performance of the system using energy and exergy analysis.

Solar drying systems capture solar radiation and convert it into thermal energy, which is then used to heat air and remove moisture from food products. The efficiency of such systems depends on several factors such as solar radiation intensity, airflow rate, collector design, and environmental conditions. Therefore, these parameters must be carefully measured and analyzed during the drying process.

In this project, a solar drying system was designed and tested to observe temperature variations, moisture removal, and overall drying performance. Experimental data were collected and analyzed using energy and exergy equations to determine the efficiency of the solar dryer. This chapter describes the components of the solar drying system, the experimental procedure, and the methods used to analyze system performance.

4.2 Description of the Solar Drying System

The solar drying system used in this project consists of several key components including a solar air collector, drying chamber, airflow system, and measuring instruments. Each of these components plays an important role in ensuring effective drying of agricultural products.

The solar air collector absorbs solar radiation and converts it into heat energy. Typically, the collector contains an absorber plate that captures solar

radiation and transfers the heat to the air flowing through the collector. The heated air is then directed into the drying chamber.

The drying chamber is the section of the system where agricultural products are placed for drying. The hot air from the solar collector enters the drying chamber and flows over the product trays. As the air moves across the product, moisture evaporates from the product and is carried away by the airflow. This process gradually reduces the moisture content of the product and increases its shelf life.

Air movement in the dryer is controlled by an airflow system. In some systems, airflow occurs naturally due to temperature differences between hot and cold air. In other systems, fans or blowers are used to force air through the dryer, which improves heat transfer and increases the drying rate.

The solar drying system used in this mini project was designed to provide a controlled drying environment while utilizing renewable solar energy. The design allows efficient heat transfer from the collector to the drying chamber and ensures proper airflow for moisture removal.

4.3 Components of the Solar Drying System

The solar drying system consists of several components that work together to achieve efficient drying.

4.3.1 Solar Air Collector

The solar air collector is the most important component of the solar dryer. Its main function is to absorb solar radiation and convert it into thermal energy. The collector typically consists of a transparent cover, absorber plate, insulation, and air channels. The transparent cover allows solar radiation to reach the absorber plate, which then transfers heat to the air flowing through the collector.

Proper insulation is provided in the collector to reduce heat losses to the surroundings. The heated air produced in the collector is then directed into the drying chamber where the drying process takes place.

4.3.2 Drying Chamber

The drying chamber is the section where agricultural products are placed for drying. The chamber contains trays or racks that hold the product. Heated air from the solar collector enters the chamber and flows across the product trays, removing moisture from the product.

The drying chamber is usually designed to ensure uniform airflow across all trays so that the product dries evenly. Proper ventilation is also necessary to remove the moist air generated during the drying process.

4.3.3 Airflow System

The airflow system controls the circulation of air through the solar dryer. Natural convection may occur due to temperature differences between hot and cold air. In forced convection systems, fans or blowers are used to circulate air through the dryer.

Forced convection generally improves the drying rate because it increases heat transfer and ensures continuous removal of moisture from the drying chamber.

4.3.4 Measuring Instruments

Various measuring instruments are used to monitor system performance during the experiment. These instruments help in observing the operating conditions of the system.

The commonly used instruments include:

- Thermometers or thermocouples for measuring air temperature at different points in the system.
- Solar radiation meters (radiometers) for measuring solar intensity.
- Hygrometers or humidity sensors for measuring moisture content in the air.
- Digital weighing balances for measuring the weight of the product before and after drying.

These measurements help determine the drying efficiency and overall system performance.

4.4 Energy Analysis

Energy analysis is used to evaluate how effectively solar energy is converted into useful heat for the drying process. In a solar drying system, solar radiation absorbed by the collector heats the air flowing through it. The heated air then transfers heat to the product in the drying chamber, which causes moisture evaporation.

Energy analysis helps determine the useful heat gain, collector efficiency, and overall system performance. The useful heat gain depends on the mass flow rate of air, the specific heat capacity of air, and the temperature difference between inlet and outlet air.

By calculating the energy efficiency of the solar collector and drying chamber, it is possible to determine how effectively the system utilizes solar energy. If a large portion of the energy is lost to the environment, improvements can be made in system design, insulation, or airflow control to enhance the performance of the solar dryer.

4.5 Exergy Analysis

While energy analysis evaluates the quantity of energy used in the system, exergy analysis evaluates the quality of energy. Exergy represents the maximum useful work that can be obtained from a system as it reaches equilibrium with its surroundings.

In solar drying systems, exergy losses occur due to temperature differences, heat transfer inefficiencies, and environmental conditions. Exergy analysis helps identify where these losses occur and suggests possible ways to minimize them.

The important parameters used in exergy analysis include:

- Exergy efficiency
- Exergy loss

- Sustainability index
- Improvement potential

These parameters provide a deeper understanding of the thermodynamic performance of the solar dryer and help identify possible methods for improving system efficiency.

4.6 Data Collection Procedure

During the experiment, data are collected at regular time intervals in order to analyze the performance of the solar drying system. The parameters measured during the experiment include:

- Solar radiation intensity
- Inlet air temperature
- Outlet air temperature
- Ambient temperature
- Moisture content of the drying product

The weight of the product is measured at different time intervals to determine the moisture loss during the drying process. These measurements help evaluate the drying rate and overall system performance.

The collected data are then used to calculate energy efficiency, exergy efficiency, and other performance parameters of the solar dryer.

4.7 Summary

This chapter described the experimental setup and methodology used to evaluate the performance of the solar drying system. The main components of the solar dryer, including the solar air collector, drying chamber, airflow system, and measuring instruments, were explained in detail.

The chapter also discussed the methods used for energy and exergy analysis, which are important tools for evaluating the thermodynamic performance of

solar drying systems. Finally, the data collection procedure used during the experiment was described.

The experimental data obtained from the solar dryer will be analyzed in the next chapter to evaluate the performance of the system in terms of drying efficiency, energy efficiency, and exergy efficiency.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the experimental study of the solar drying system and discusses the performance of the system based on the collected data. The purpose of this chapter is to analyze the drying behavior of the agricultural product and evaluate the thermodynamic performance of the solar dryer using energy and exergy analysis.

During the experiment, several parameters such as solar radiation intensity, inlet air temperature, outlet air temperature, ambient temperature, and moisture content of the product were measured at fixed time intervals. These parameters were used to determine the drying rate, energy efficiency, and exergy efficiency of the solar drying system.

The results obtained from the experimental system help in understanding how effectively the solar dryer utilizes solar energy and how the system performs under different environmental conditions.

5.2 Temperature Variation During Drying

Temperature is an important factor in the drying process because it directly affects the rate at which moisture evaporates from the product. During the experiment, the temperatures at different points in the solar drying system were recorded, including the ambient temperature, inlet air temperature, and outlet air temperature.

The results showed that the air temperature inside the solar collector increased significantly compared with the ambient temperature due to the absorption of solar radiation by the absorber plate. As solar radiation increased during the daytime, the collector temperature also increased, resulting in higher air temperatures entering the drying chamber.

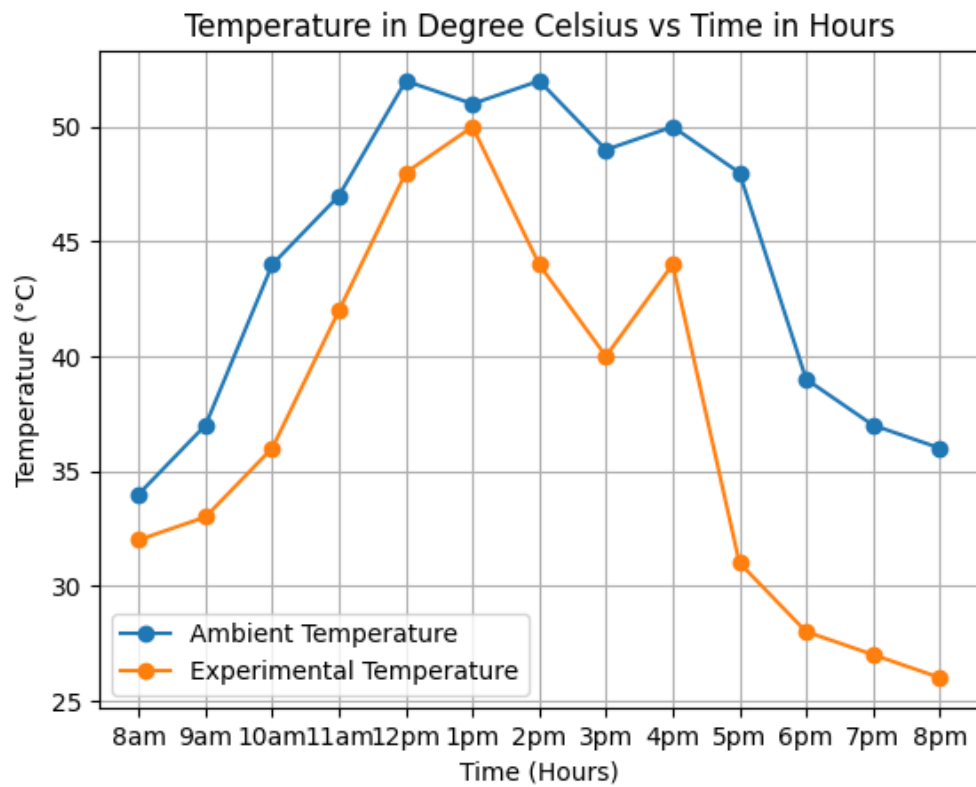


Figure 5.1: Ambient and Experimental Temperatures of Moringa

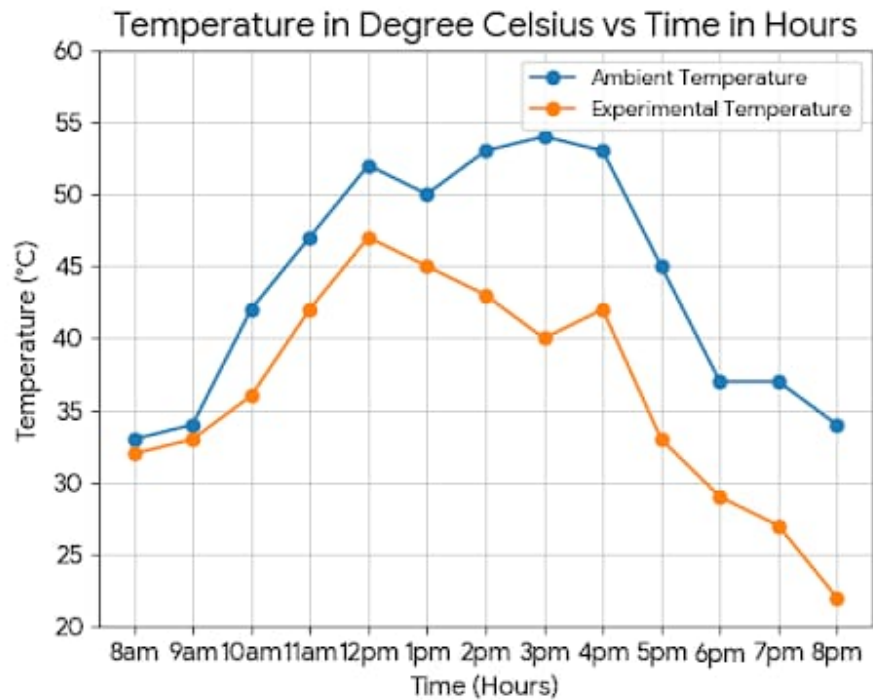


Figure 5.2: Ambient and Experimental Temperatures of Plums

The outlet air temperature from the drying chamber was generally lower than the inlet air temperature because part of the thermal energy was used for evaporating moisture from the product. This temperature difference indicates the transfer of heat from the drying air to the product.

The highest temperatures were observed during midday when solar radiation intensity was at its peak. During this period, the drying process was most effective because higher temperatures accelerated the evaporation of moisture from the product.

5.3 Weight Variation During Drying

Weight variations of the moringa leaves during the solar drying process were evaluated by recording the initial and the final mass of the samples at different time intervals throughout the experiment. Weight loss represents drying of the leaves due to the combined effect of solar radiation and heated air inside the drying chamber. As the solar drying process progressed, it was observed that the weight of the samples kept decreasing at every stage, which signifies a continued mass loss during the process, pointing at ongoing vaporization of moisture from the samples. The highest and lowest weight losses recorded during the experiment were 36 g and 4 g, respectively, attributed to variations in drying parameters, including solar intensity, temperature, and airflow rate. The weight loss increments clearly indicate that the solar drying system is effective enough in extracting moisture from moringa leave within a given time

Energy analysis was performed to evaluate how efficiently solar energy was converted into useful heat for the drying process. The energy efficiency of the solar dryer depends on the amount of solar radiation received by the collector and the useful heat transferred to the drying air.

The results showed that the solar air collector effectively absorbed solar radiation and converted it into thermal energy. The heated air produced in the collector provided sufficient heat to the drying chamber for moisture evaporation.

However, some energy losses occurred due to heat transfer to the surround-

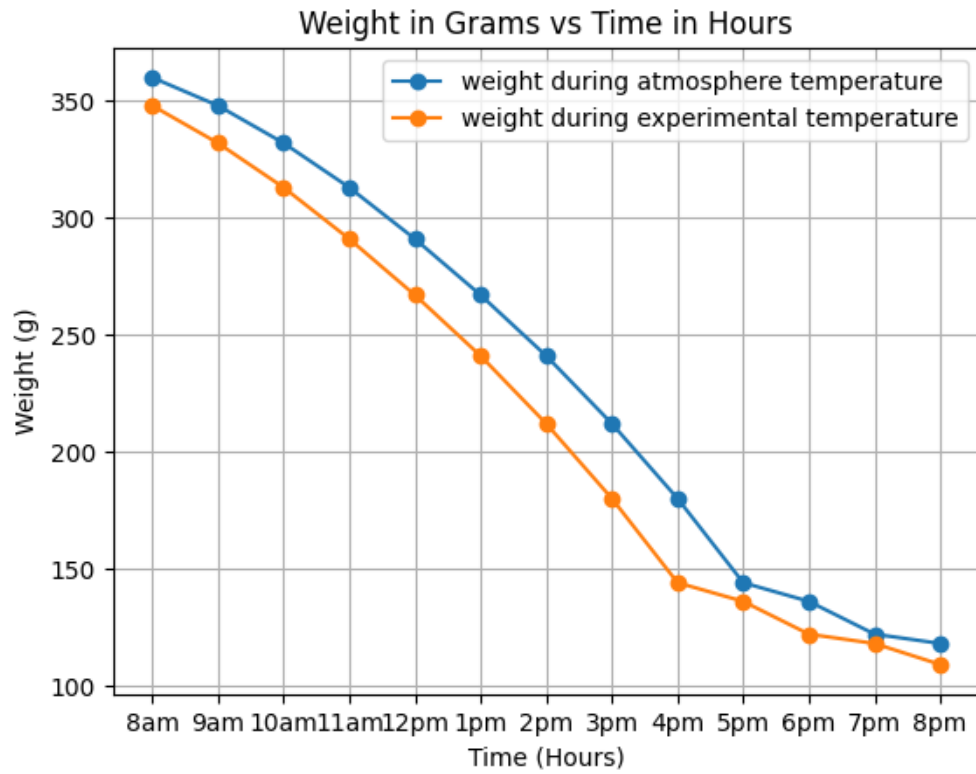


Figure 5.3: Weight during atmosphere and experimental temperatures of moringa

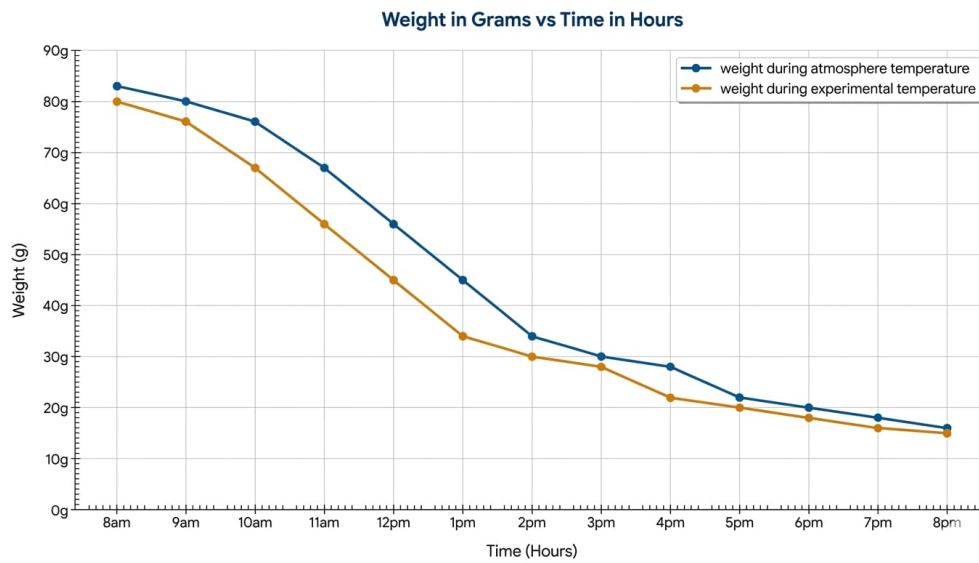


Figure 5.4: Weight during atmosphere and experimental temperatures of Plums

ings, air leakage, and temperature differences between system components. Despite these losses, the solar dryer demonstrated good energy utilization during the experiment.

Energy efficiency was found to be higher during periods of strong solar radiation because more solar energy was available to heat the air. During cloudy conditions or during early morning and evening hours, the efficiency decreased due to lower solar radiation intensity.

Table 5.1: Experimental Data of Moringa in Solar Drying Process

Time	Mass Flow	Inlet Temp	Outlet Temp	DC In Temp	DC Out Temp	Area (m ²)	Solar (W/m ²)	Initial Mass	Final Mass	Outlet Temp	Ambient Temp	Latent (kJ/kg)	Heat (W)	Collector Eff (%)	Dryer Eff (%)
8.00	0.047	26.35	35	34	32	3	430	360	348	35	28	2260	408.58	31.67	-191.78
9.00	0.047	27.35	39	37	33	3	450	348	332	39	29	2260	550.28	40.76	51.17
10.00	0.047	32.2	43	44	36	3	560	332	313	46	32	2260	510.13	30.36	215.27
11.00	0.047	35	49	47	42	3	680	313	291	51	34	2260	661.29	32.41	91.96
12.00	0.05	38	51	52	48	3	780	291	267	55	36	2260	653.25	27.91	64.66
13.00	0.05	40	50	51	50	3	800	267	241	60	40	2260	502.50	20.93	-64.93
14.00	0.06	42	53	52	44	3	870	241	212	61	42	2260	663.30	25.41	252.70
15.00	0.05	36	51	49	40	3	720	212	180	55	39.28	2260	753.75	34.89	288.05
16.00	0.047	32	49	50	44	3	634	180	144	50	34.87	2260	802.99	42.21	169.01
17.00	0.047	28	34	48	31	3	580	144	136	45	28.86	2260	283.41	16.28	418.47
18.00	0.047	29	36	39	28	3	460	136	122	41	26.52	2260	330.64	23.95	372.39
19.00	0.047	28	35	37	27	3	300	122	118	39	25.72	2260	330.64	36.73	94.254
20.00	0.047	27	34	36	26	3	250	118	109	38	24.35	2260	330.64	44.08	212.04

Table 5.2: Experimental Data of Plums in Solar Drying Process

Time	Mass Flow	Inlet Temp	Outlet Temp	DC In Temp	DC Out Temp	Area (m ²)	Solar (W/m ²)	Initial Mass	Final Mass	Outlet Temp	Ambient Temp	Latent (kJ/kg)	Heat (W)	Collector Eff (%)	Dryer Eff (%)
08:00	0.047	24	29	33	32	3	430	83	80	35	28	2260	236.175	18.308	-41.4735
09:00	0.047	28	31	34	33	3	450	80	76	39	29	2260	141.705	10.497	-35.4257
10:00	0.047	32	43	42	36	3	560	76	67	46	32	2260	519.585	30.928	65.3021
11:00	0.047	36	50	47	42	3	680	67	56	51	34	2260	661.29	32.416	45.9934
12:00	0.05	38	51	52	47	3	780	56	45	55	36	2260	653.25	27.917	46.5595
13:00	0.05	39	52	50	45	3	800	45	34	60	40	2260	653.25	27.219	46.5595
14:00	0.06	42	53	53	43	3	870	34	30	61	42	2260	663.3	25.414	46.9772
15:00	0.05	37	55	54	40	3	720	30	28	55	39.28	2260	904.5	41.875	26.1139
16:00	0.047	33	50	53	42	3	634	28	22	50	34.87	2260	802.995	42.218	65.7165
17:00	0.047	28	34	45	33	3	580	22	20	45	28.86	2260	283.41	16.288	69.1578
18:00	0.047	28	35	37	29	3	460	20	18	41	26.53	2260	330.645	23.960	34.962
19:00	0.047	28	35	37	27	3	300	18	16	39	25.73	2260	330.645	36.738	47.120
20:00	0.047	27	34	34	22	3	250	16	15	38	24.35	2260	330.645	44.086	29.639

5.4 Exergy Analysis Results

Exergy analysis was conducted to evaluate the quality of energy used in the solar drying system and to identify thermodynamic losses. Unlike energy analysis, exergy analysis provides information about the maximum useful work that can be obtained from the energy available in the system.

The results indicated that exergy losses occurred mainly in the solar air collector and during the heat transfer process between the drying air and the product. These losses were caused by temperature differences, heat transfer inefficiencies, and environmental conditions.

Exergy efficiency was found to be lower than energy efficiency because not all the available energy could be converted into useful work. However, the

exergy analysis provided valuable insights into where improvements could be made in the system design.

Parameters such as exergy efficiency, exergy loss, sustainability index, and improvement potential were evaluated to better understand the thermodynamic behavior of the system. These parameters help identify opportunities to improve the performance of solar drying systems.

5.5 Discussion of System Performance

The experimental results demonstrate that the solar drying system is capable of effectively utilizing solar energy for drying agricultural products. The solar air collector successfully increased the air temperature, which enhanced moisture evaporation and improved drying performance.

The results also showed that the drying rate and system efficiency are strongly influenced by environmental conditions such as solar radiation intensity, ambient temperature, and airflow rate. Proper system design and insulation can help reduce heat losses and improve system efficiency.

Energy and exergy analyses provided important insights into the thermodynamic performance of the solar dryer. While energy analysis indicated how much solar energy was utilized, exergy analysis revealed the quality of energy and the locations where energy losses occurred.

By improving collector design, insulation, and airflow control, it is possible to reduce energy losses and enhance the overall efficiency of solar drying systems.

5.6 Summary

This chapter presented the results and discussion of the experimental study conducted on the solar drying system. The temperature variation, moisture removal, and the energy and exergy efficiency of the system were analyzed.

The results showed that the solar dryer effectively utilized solar energy for drying agricultural products and provided a sustainable alternative to conventional drying methods. Energy and exergy analyses were used to evaluate



Figure 5.5: Fully dried moringa

the thermodynamic performance of the system and to identify possible areas for improvement.

The findings of this study demonstrate the potential of solar drying systems as efficient and environmentally friendly technologies for food preservation.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This project focused on the energy and exergy analysis of a solar drying system in order to evaluate its performance and efficiency in drying agricultural products. Solar drying systems provide an environmentally friendly and sustainable alternative to conventional drying methods that rely on fossil fuels or electricity. The main objective of this study was to analyze how effectively solar energy can be utilized for drying and to understand the thermodynamic behavior of the system.

From the experimental investigation, it was observed that the solar drying system was capable of converting solar radiation into useful thermal energy for the drying process. The solar air collector played an important role in absorbing solar radiation and increasing the temperature of the air entering the drying chamber. The heated air then removed moisture from the product, resulting in a gradual reduction of moisture content and improved preservation of the agricultural product.

The temperature measurements during the experiment showed that the collector temperature was higher than the ambient temperature due to the absorption of solar radiation. The inlet air temperature to the drying chamber increased significantly during periods of high solar radiation, which improved the drying performance. The outlet air temperature was slightly lower than the inlet air temperature because part of the thermal energy was used for moisture evaporation.

The moisture removal analysis indicated that the drying rate was highest during the initial stages of the drying process. As drying continued, the rate gradually decreased because the remaining moisture was bound within the internal structure of the product. Environmental factors such as solar radiation

intensity, ambient temperature, and airflow rate were found to significantly influence the drying performance.

Energy analysis revealed that the solar drying system was able to effectively utilize solar energy for heating the air and supporting the drying process. Although some heat losses occurred due to heat transfer to the surroundings and system inefficiencies, the overall performance of the solar dryer was satisfactory.

Exergy analysis provided deeper insights into the thermodynamic behavior of the system by identifying energy losses and irreversibilities. The results indicated that exergy losses occurred mainly in the solar air collector and during heat transfer between the drying air and the product. Despite these losses, the solar drying system demonstrated reasonable thermodynamic performance.

Overall, the results of this project demonstrate that solar drying systems can be effectively used for drying agricultural products while utilizing renewable solar energy. The integration of energy and exergy analysis provides valuable information for evaluating system efficiency and identifying areas for improvement.

6.2 Future Scope of Work

Although the present study successfully evaluated the performance of the solar drying system, there are several areas where further research and improvements can be made to enhance the system performance.

One possible area of improvement is the design of the solar air collector. Improving the absorber plate material and increasing the collector surface area could enhance the absorption of solar radiation and increase the temperature of the drying air. Better insulation materials could also be used to minimize heat losses from the collector and drying chamber.

Another potential improvement is the integration of thermal energy storage systems such as phase change materials (PCM). Thermal energy storage would allow the solar dryer to maintain stable temperatures even during periods of low solar radiation or after sunset. This would help extend the drying period and improve the overall efficiency of the system.

The use of forced convection airflow systems powered by photovoltaic panels could also improve drying performance. Controlled airflow can enhance heat transfer and accelerate moisture removal from the product. Combining solar thermal energy with solar photovoltaic systems would make the drying system more sustainable and energy efficient.

Further research could also focus on drying different types of agricultural products and analyzing how different drying conditions affect product quality, drying time, and system efficiency. Studies can also be conducted to optimize system parameters such as airflow rate, collector angle, and drying chamber design.

In addition, advanced monitoring techniques and sensors can be incorporated into the system to improve data collection and system control. These technologies could help optimize the drying process and improve the accuracy of energy and exergy analysis.

Finally, future work could explore the development of hybrid solar drying systems that combine solar energy with auxiliary heating sources to ensure continuous drying under varying weather conditions.

In conclusion, the results of this project show that solar drying systems have strong potential for sustainable food preservation. With further improvements in system design, thermal storage, and airflow control, solar dryers can become an even more efficient and reliable technology for agricultural applications.

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